Project Management

Group Project Assignment Phase 1

Project Topic: FitSpot Gym Management System & Mobile App

Group 1 Part Time

Group Members

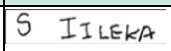
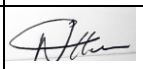
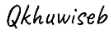
#	Full Name	Student Number	Signature
1	Selma Iileka	223102415	
2	Modestus Kandowa	222071567	
3	Titus Shaangelo	222084774	
4	Elise Mwandingi	218110510	
5	Naikuti Nambinga	224016288	
6	Matengu Mukowa	223119199	
7	Jefta Jesaya	223089788	
8	Simeon Shileka	218053444	
9	David Hailume	223096709	*David Hailume*
10	Hermaine Kharugas	224001833	
11	Quinton Khuwiseb	224031503	

TABLE OF CONTENT

Project team list.....2
Kick-off meeting agenda.....3
Project topic justification.....5
Team contract.....5
Business case.....6
Project charter.....10
Stakeholder register.....15
Reference list.....17

Project Team List

Student Number	Name & Surname	Role / Position
218053444	Simeon T. Shileka	Project Manager / System Architect
223089788	Jefta Jesaya	Project Coordinator
223102415	Selma Iileka	UI/UX Developer
224001833	Hermaine Kharugas	Backend Developer
224031503	Quinton Khuwiseb	Mobile Frontend Developer
222084774	Titus Shaangelo	Quality Assurance (QA) Officer
222071567	Modestus Kandowa	Requirements Engineer
218110510	Elise Mwandingi	Lead Developer
224016288	Naikuti Nambinga	Development and Operations / Systems engineer
223119199	Matengu Denzel	Communications & Marketing Officer
223096709	David Hailume	Project Administrator

Kick-Off Meeting
26 August 2026

Project Name: FitSpot Gym Management System & Mobile App

Meeting Objective: Get the project off to an effective start by introducing key stakeholders, reviewing project goals, and discussing future plans.

Agenda:

1. **Introductions of Attendees**
    - Introducing all group members
    - Confirm Project Manager and team roles
2. **Discuss the Possible IT Projects**
    - Presenting the proposed projects
    - Briefly explain the purpose of each project
3. **Review Project Backgrounds**
    - Identify the problem or opportunity addressed by each project
    - Discuss the intended users and organizations/businesses that would benefit
4. **Evaluate the Proposed Projects**
    - Alignment with business strategies
    - Potential financial benefits
    - Other potential benefits
    - Feasibility and overall project value
5. **Select the Preferred IT Project**
    - Compare the proposed projects
    - Discuss advantages and disadvantages
    - Agree on the project to be developed
6. **Brainstorm and Define the Selected Project**
    - Discuss the project's key objectives
    - Identify potential stakeholders
    - Establish the initial project scope
7. **Assign Team Roles and Responsibilities**
    - Confirm responsibilities for each team member
    - Assign project-related tasks
8. **Assign Project Documentation**
    - Business Case
    - Project Charter
    - Stakeholder Register
    - Project Scope
9. **Discuss Project Risks, Assumptions and Constraints**
    - Identify initial risks
    - Establish project assumptions
    - Identify project constraints
10. **Discuss Project Scope, Time and Cost Goals**
    - Define initial scope boundaries
    - Establish preliminary project timeline
    - Discuss estimated project costs and resources

11. Confirm Communication and Reporting Methods

- Communication platforms
- Progress reporting

12. Action Items and Next Steps

- Summarize decisions made
- Assign outstanding tasks
- Set deadlines
- Confirm date of next meeting

Action Item	Assigned To	Due Date
Approves the project objectives and scope.	Project Sponsor	26 August 2026
Confirm project and governance structures.	Project Manager	26 August 2026
Identify and classify key stakeholders.	Project Manager	26 August 2026
Define and establish project scope.	Product Owner & Project manager	26 August 2026
Define and circulate the initial project charter and stakeholders register.	Project Manager	28 August 2026
Develop initial project schedule.	Project manager	28 August 2026
Review Phase 1 documentation.	Entire team	01 September 2026
Schedule requirements gathering workshops and meetings.	Requirements Engineer	05 September 2026
Prepare initial UI/UX design wireframes for stakeholders to review.	UI/UX Designer	10 September 2026
Design initial database schema and draft the system architecture.	Backend Developer	10 September 2026
Develop a test plan and criteria for verifying project deliverables.	Quality Assurance (QA) Lead	15 September 2026

Date and time of next meeting: 06 September 2026

Project topic justification

After considering three possible projects, the Gym Management System, Campus Lost and Found System, and Domestic Worker Booking System, the group selected the Gym Management System (FitSpot Gym Management System & Mobile App).

The Gym Management System was chosen because it provides a good balance between financial benefits, practical usefulness, higher rate of customer satisfaction, and technical feasibility. It will generate revenue through gym memberships, and premium subscription services, while reducing administrative overhead and improving equipment management and maintenance.

The Campus Lost and Found System would provide useful social and operational benefits, but it has limited opportunities for direct financial benefits. The Domestic Worker Booking System has strong commercial potential, but it would be more complex due to requirements such as worker verification, customer safety and dispute management.

Overall, the Gym Management System was considered the most suitable project to embark on, as it has clear financial and operational benefits while remaining realistic and manageable for the project team.

Team Contract

Section	**Agreement**
Purpose	All members commit to completing the FitSpot Gym Management System & Mobile App project successfully.
Roles & Responsibilities	Each member agrees to fulfill their assigned role as listed in the Team List.
Communication	The team will have weekly meetings, use WhatsApp for daily updates, and rely on emails for official communication.
Decision-Making	Consensus-based; if unresolved, Project Manager has final say after consulting Project Coordinator and Lead Developer.
Attendance & Participation	Members must attend meetings unless excused; active participation in tasks is mandatory.
Conflict Resolution	Issues discussed openly; unresolved conflicts escalated to Project Manager and Coordinator.
Accountability	Members complete tasks by deadlines; failure without valid reason may lead to redistribution of tasks.

BUSINESS CASE

Project Name: FitSpot Gym Management System & Mobile App

1.0 Introduction / Background

Most gyms still rely on manual, paper-based methods to manage member registration, payments, attendance, trainer information, and equipment records. These processes are slow, inconsistent, and prone to errors, which can result in lost records and frustrated staff and members.

The FitSpot Gym Management System & Mobile App will replace these manual processes with a centralized digital platform that provides access to gym services, schedules, and facility information on the gym members' phone. It will enable staff to register members, process secure payments, track equipment maintenance, and monitor subscription trends via a comprehensive dashboard. Furthermore, the app allows membership self-registration, booking specialized group classes and personal training, streaming video workouts and viewing live gym zone utilization analytics to avoid peak hours and effectively plan their workout schedules.

2.0 Business Objective

The main objective of the Gym Management System is to modernize the gym's daily operations and improve the quality of service delivered to members through a single digital platform. The project will:

- Reduce reliance on paper-based and manual record-keeping.
- Make member registration faster and simpler.
- Maintain accurate membership and payment records.
- Give staff a reliable way to track member attendance.
- Support the scheduling and management of trainers.
- Improve oversight of gym equipment and its maintenance.
- Reduce errors in billing and membership information.
- Strengthen communication between gym members and staff.

3.0 Current Situation and Problem/Opportunity Statement

Current Situation:

The gym currently depends on manual methods such as logbooks, spreadsheets, and separate records to manage members, payments, attendance, and other daily activities.

Problem:

- Member records are frequently misplaced or difficult to locate.
- Manual registration takes up unnecessary time.
- Payment records are vulnerable to human error.
- Membership expiry dates are difficult to track.
- Attendance is monitored inconsistently.
- Equipment records are not always kept up to date.
- Staff cannot always access information when they need it.

Opportunity:

A centralized Gym Management System would bring all of this information into one platform with advanced features, making gym operations faster, better organized, and easier to monitor.

4.0 Critical Assumptions and Constraints

Assumptions:

- Gym staff will be willing to adopt the new system.
- Members will provide accurate personal information.
- The gym will have access to the required computers and internet.
- Sufficient funding and technical expertise will be available.
- Management will support the system's implementation.

Constraints:

- The project must be delivered within a limited budget.
- The system must be completed within the agreed project period.
- The project team has limited time and resources.
- Staff will need training before the system goes live.
- The system depends on stable electricity, computers, and network access.

5.0 Analysis of Options and Recommendation

Option 1: Continue with the Current Manual System

Keeping the existing logbooks, paper records, and spreadsheets has a low upfront cost, but operations remain slow, disorganized, and prone to error.

Option 2: Use Separate Software for Different Functions

Adopting different applications for membership, payments, attendance, and equipment could improve individual tasks, but would leave information scattered across multiple systems and harder to manage as a whole.

Option 3: Develop a Gym Management System — Recommended

The recommended option is to build one centralized Gym Management System that brings all core gym functions together. This approach improves operational efficiency, reduces administrative errors, makes information accessible, improves gym member satisfaction and retention, and increases onboarding of new members.

6.0 Preliminary Project Requirements

The system should include:

- Member registration, profile management, and self-service subscription upgrades, downgrades, and cancellations.
- Secure payment processing and history generation.
- Centralized, real-time access to gym services, schedules, and facility information.
- Push notifications for facility announcements, events, and motivational prompts.
- Booking for specialized clubs, group classes, and personal training.
- Subscription-based video workout streaming for gym and at-home training.
- Subscription-based custom at-home workout routine generation based on available equipment.
- Subscription-based access to personalized nutritional guides and meal tracking.
- Real-time zone utilization analytics (cardio and weight zones) to avoid peak hours.
- Administrative dashboards to monitor subscriptions, usage trends, and operations.
- Gym equipment inventory and maintenance tracking.
- Secure data storage, user authentication, and role-based access control.

- Reliable access and performance under intermittent internet conditions.

7.0 Budget Estimate and Financial Analysis

Estimated Budget Categories:

- Platform development and infrastructure
- Licensing, hosting and security
- Payment gateway integration
- Communications APIs
- Marketing and awareness campaigns
- Training and capacity building
- Maintenance and support
- Contingency reserve

Financial Analysis:

The project will require an initial investment for platform development, third-party cloud and payment gateway integration and comprehensive testing. Over the long term, the return on investment (ROI) will be greater than the invested capital from digital subscriptions (video training workouts, nutritional guides) and streamlined operations. Administrative overhead will drop significantly due to automated member registration, centralized booking and digital equipment maintenance tracking. Furthermore, increased customer engagement via push notifications, real-time gym zone analytics, and reliable performance will increase customer satisfaction rates and, in return, long-term member retention. This will increase the gym member base and revenue generation. Coupled with accurate and secure payment protocols and analytics tracking will aid successful gym management and improve overall profitability.

8.0 Schedule Estimate

- Requirements gathering and stakeholder consultations — 2 weeks
- System architecture design and UI/UX wireframing — 2 weeks
- Database schema design and local payment mapping — 2 weeks
- User authentication, profiles, and API booking — 3 weeks
- Administrative dashboards and equipment maintenance tracking — 3 weeks
- Core mobile app framework and initial feature integration — 3 weeks
- Internal prototype testing with gym operators — 2 weeks
- Payment gateway, push notification, and SMS integrations — 2 weeks
- Video streaming, nutrition tracking and gym zones utilization analytics — 3 weeks
- Offline syncing for intermittent internet connections and quality assurance testing — 3 weeks
- Deployment, user training, and commercial platform launch — 2 weeks
- Closure and Handover — 1 week

Total project duration: 7 months

9.0 Potential Risks

The project may face the following risks:

- Resistance from gym staff to using the new system.

- Lack of sufficient funding.
- Delays in system development.
- Technical problems during implementation.
- Loss of data if proper backups are not maintained.
- Cybersecurity threats.
- Incorrect information entered by users.
- System downtime caused by network or electricity problems.

10.0 Exhibits

Exhibit A: Financial Analysis

- Comparison of the cost of the current manual system and the proposed digital system.
- Estimated cost savings from reducing paper and manual work.
- Possible increase in revenue through better membership and payment tracking.
- Projected reduction in maintenance and operational costs.

Financial Analysis / Net Present Value (NPV) Table

Financial Metric / Year (N$)	**Year 0**	**Year 1**	**Year 2**	**Year 3**	**Year 4**	**Year 5**	**Total**
Projected Revenue		280,000	320,000	360,000	400,000	450,000	1,810,000
Initial Investment	(565,000)						(565,000)
Annual Operating Costs		(100,000)	(100,000)	(100,000)	(100,000)	(100,000)	(500,000)
Net Cash Flow	**(565,000)**	**180,000**	**220,000**	**260,000**	**300,000**	**350,000**	**745,000**
Discount Factor (10%)	1.00	0.91	0.83	0.75	0.68	0.62	
Present Value (PV)	**(565,000)**	**163,800**	**182,600**	**195,000**	**204,000**	**217,000**	**397,400 (NPV)**

Return on Investment (ROI): 131.86%

Project Charter

Project Title: FitSpot Gym Management System & Mobile App

Project Start Date: 01 September 2026 **Projected Finish Date:** 01 April 2027

Budget Information:

- Platform development and infrastructure: **N$ 300,000**
- Licensing, hosting and security: **N$ 45,000** (annually)
- Payment gateway integration: **N$ 12,000**
- Communications APIs: **N$ 20,000**
- Marketing and awareness campaigns: **N$ 40,000**
- Training and capacity building: **N$ 18,000**
- Maintenance and support: **N$ 55,000** (annually)
- Contingency reserve: **N$ 75,000**

Estimated initial budget: **N$ 565,000**

Project Manager:

Simeon Shileka
+264 85 7739177
shilekastn@gmail.com

Project Objectives:

- Provide centralized and real-time access to gym services, schedules, and facility information.
- Strengthen communication by using push notifications for general facility announcements, community events, and motivational prompts to keep members active.
- Enable seamless member registration and secure payments.
- Allow users to digitally upgrade, downgrade, or cancel their subscription on demand.
- Facilitate sign-ups for specialized gym clubs, group classes, and personal training.
- Provide subscription-based video training workouts tailored for both gym and at-home training.
- Generate customized workout routines based on the user's available equipment for at-home training.
- Provide subscription-based access to personalized nutritional guides and calorie tracking from a certified nutritionist.
- Display real-time zone utilization analytics (e.g., cardio, weight zones), to help members avoid peak hours.
- Give administrators comprehensive dashboards to monitor subscriptions and usage trends.
- Improve oversight of gym equipment and its maintenance.
- Implement robust data privacy and secure user authentication and payment protocols.
- Design an app to perform reliably under intermittent internet conditions.

Success Criteria:

- **User adoption:** Achieve at least 60% of active gym members registering and actively using the app for bookings or check-ins within the first three months of launch.
- **Business engagement:** 80% of gym staff and personal trainers consistently utilizing the management dashboard for client scheduling, tracking, and operational planning.
- **Service integration:** Successfully process 95% of digital membership payments and class bookings without failure through integrated local payment and notification channels.
- **Operational efficiency:** Reduce administrative check-in bottlenecks by at least 40% through self-service digital onboarding and QR-code access verifications.
- **System reliability:** Maintain a 99.5% application uptime, with fast loading and graceful fallback handling during intermittent local network fluctuations.
- **Financial accuracy:** Achieve 100% data reconciliation between automated in-app payment logs and gym bank statements to eliminate revenue leakage.
- **Data privacy and compliance:** Zero security breaches or unauthorized disclosures for user personal identifiable information (PII) and financial records throughout the project lifecycle.
- **Stakeholder satisfaction:** Attain an average satisfaction rating of 4.2 out of 5 stars from both gym members (via app reviews) and administrators (via usability feedback).

Approach:

Phase 1 - Research and architecture design (Months 1-2):

- Conduct stakeholder consultations with gym owners, trainers, and members.
- Gather system requirements, map out database schemas, and finalize UI/UX wireframes and local payment parameters.

Phase 2 – Core development and prototype (Months 3-5):

- Build core platform features (such as user authentication, administrative dashboards, membership profiles, and booking APIs) and run prototype tests with select gym services.

Phase 3 – Implementation and integration (Month 6-7):

- Integrate advanced modules (local payment gateways, push notifications, crowd analytics, and video streaming).
- Launch the platform commercially and execute targeted marketing and awareness campaigns.

Phase 4 – Monitoring and continuous improvement (Ongoing):

- Conduct rigorous quality assurance, collect user feedback, track system analytics, and roll out new features and optimizations.

Roles and Responsibilities

Name and Surname	***Role***	***Position***	***Contact Information***
Simeon T. Shileka	* Lead project initiation. * Planning and scheduling. * Technical architecture oversight. * Resource allocation. * Monitoring and closure.	Project manager / System architect	+264 85 7739177 shilekastn@gmail.com
Jefta Jesaya	* Define project vision. * Set objectives. * Maximize business value. * Prioritize deliverables. * Bridge stakeholders and team.	Project coordinator	+264 81 663 0735 jeftajesaya04@gmail.com
Selma Iileka	* Design user interfaces. * Build wireframes and prototypes. * Ensure usability and accessibility. * Maintain aesthetic standards. * Coordinates user flows with developers.	UI/UX developer	+264 81 761 7816 kittyselma9@gmail.com
Hermaine Kharugas	* Implement core system logic. * Develop and secure APIs. * Design database structures. * Ensure system performance. * Scale backend architecture.	Backend developer	+264 81 783 6689 kharugashermaine@gmail.com

Quinton Khuwiseb	* Develop front-end mobile UI. * Integrate APIs with client views. * Manage local state handling. * Optimize cross platform performance. * Test responsive layouts.	Mobile frontend developer	+264 81 564 9939 Khuwisebquinton@gmail.com
Titus Shaangelo	* Develop test plans and cases. * Execute functional and regression tests. * Identify and log software defects. * Verify requirements. * Coordinate bug fixes with developers.	Quality assurance (QA) officer	+264 81 487 1520 sp7titus@gmail.com
Modestus Kandowa	* Gather and document requirements. * Analyze functional specifications. * Manage scope change requests. * Bridge technical and business terms. * Validate deliverables against specifications.	Requirements engineer	+264 85 793 8168 billykandowa@gmail.com
Elise Mwandingi	* Provide technical leadership. * Supervise development tasks. * Enforce coding standards. * Review pull requests and code.	Lead developer	+264 81 225 5433 elisemegy@gmail.com

	* Mentor junior team members.		
Naikuti Nambinga	* Configure cloud hosting environments. * Manage CI/CD deployment pipelines. * Handle database backups and security. * Monitor system uptime and logs. * Optimize network infrastructure.	Development and Operations / Systems engineer	+264 81 228 7235 naikutijohn@gmail.com
Matengu Denzel	* Plan marketing and awareness campaigns. * Coordinate stakeholder outreach. * Draft user documentation and manuals. * Manage launch communications. * Gather user feedback post-launch.	Communications and marketing officer	+264 81 766 6342 mukowa8@gmail.com
David Hailume	* Track project budget and expenses. * Manage risk assessment logs. * Maintain meeting minutes and schedules. * Handle procurement and supplier relations (e.g., software licenses, hardware, and cost tracking). * Assist with project reporting.	Project administrator	+264 81 825 8599 hailumedavid42@gmail.com

Stakeholder Register

Project name: FitSpot Gym Management System & Mobile App

Name	Position	Internal/ External	Project Role	Contact Information
Simeon T. Shileka	Project manager / System architect	Internal	* Oversee overall architecture and technical workflows. * Coordinate multi-platform deployment and integration. * Supervise database modeling and backend security. * Ensure alignment with project technical specification.	+264 85 7739177 shilekastn@gmail.com
Jefta Jesaya	Project coordinator	Internal	* Formulate the core platform vision and strategic objectives. * Maximize commercial and operational value for gym stakeholders. * Prioritize feature releases and manage product backlog. * Serve as primary liaison between external gym partners and the dev team.	+264 81 663 0735 jeftajesaya04@gmail.com
Modestus Kandowa	Requirement engineer	Internal	* Gather and document system requirements from gym operators. * Analyze functional and non-functional specifications. * Manage scope change requests throughout the development lifecycle. * Bridge technical and business terms for team alignment.	+264 85 793 8168 billykandowa@gmail.com
David Hailume	Project administrator	Internal	* Maintain project documentation, meeting minutes, and repository logs.	+264 81 825 8599 hailumedavid42@gmail.com

			* Track resource allocation, schedules, and administrative tasks. * Assist with communication across teams and report generation.	
Gym owner/ Executive board	Executive sponsor	External	* Provide overall project funding and resource authorization. * Review and approve major project milestones and deliverables. * Align system deliverables with business goals and compliance requirements.	
Lead gym instructor	End-user representative	External	* Provide domain expertise on class scheduling and training workflows. * Participate in user acceptance testing for the mobile app. * Gather feedback from trainers and gym members on UI usability.	
Cloud and payment gateway vendor	Third-party service provider	External	* Supply secure API endpoints for subscription and debit order processing. * Maintain hosting infrastructure uptime and database server availability. * Provide technical support for payment integrations.	

Reference list

1. Schwalbe, K. (2014). Information technology project management (7th ed.). Course Technology, Cengage Learning.